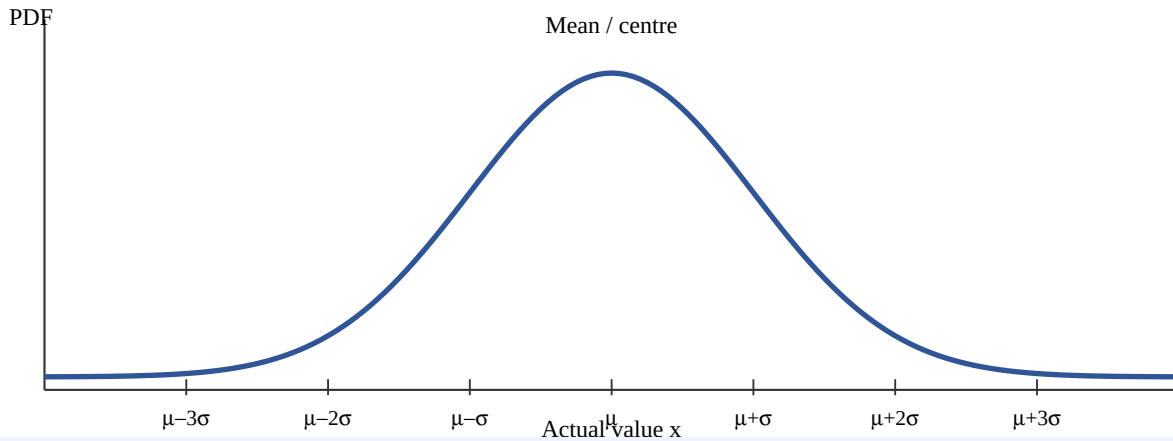


Normal Distribution & Standard Normal Distribution

Study notes: PMF vs PDF, continuous probability, normal curve, PDF formula, 68–95–99.7 rule, standard normal distribution, and z-scores.



Graph note: The normal curve approaches the x-axis in both tails but does not touch it.

Core idea: PMF gives probabilities for discrete values. PDF describes density for continuous variables. A normal distribution is a bell-shaped continuous distribution. The standard normal is the special case $\mu = 0$ and $\sigma = 1$.

1. PMF vs PDF

PMF — Probability Mass Function

A **discrete** random variable has countable possible values, such as the number on a die or the number of defective items.

$$\text{PMF: } P(X = x)$$

For a discrete variable, the PMF can give a nonzero probability to an exact possible value. Example: for a fair die, $P(X = 4) = 1/6$.

PDF — Probability Density Function

A **continuous** random variable can take values throughout an interval, such as height, weight, time, temperature, or measurement error.

A PDF gives **probability density**, not the probability of an exact point.

$$\text{For a continuous variable: } P(X = x) = 0$$

This does not mean x is impossible. A single exact point has zero width. Probabilities are assigned to **ranges**.

Clear example: If X is a person's continuous height, $P(X = 170 \text{ cm}) = 0$. Instead, ask $P(169 \text{ cm} < X < 171 \text{ cm})$, which can be positive.

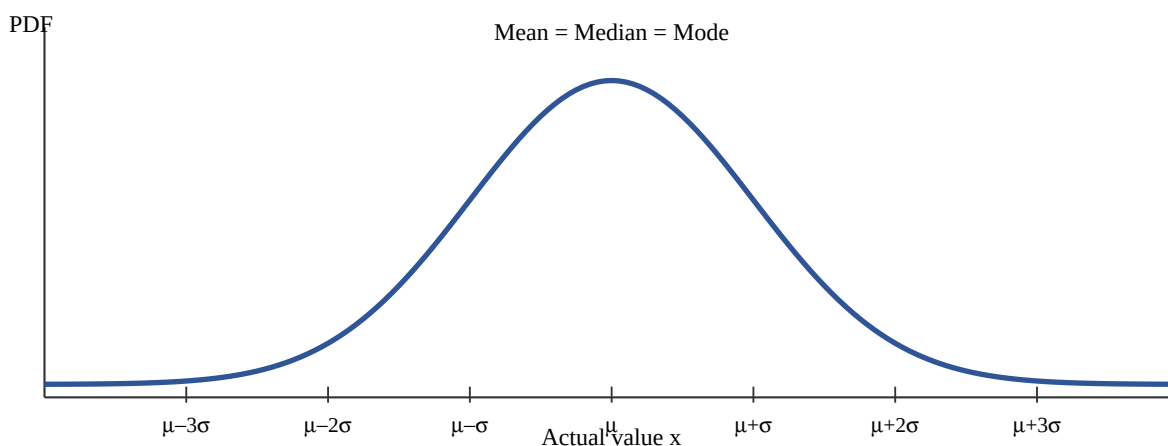
If a device reports height to the nearest centimetre, a recorded '170 cm' may represent an underlying interval such as $169.5 \leq X < 170.5$.

Feature	PMF	PDF
Variable	Discrete	Continuous
Exact point	$P(X=x)$ can be > 0	$P(X=x)=0$
What it represents	Probability mass	Probability density
Range probability	Add point probabilities	Area under the density curve

2. Normal Distribution

A **normal distribution** is a continuous probability distribution with a bell-shaped, symmetric probability-density curve. It is determined by its mean μ and standard deviation σ .

$$X \sim N(\mu, \sigma^2)$$



Graph reading: x-axis = actual values of X. y-axis = probability density. The curve approaches, but never touches, the x-axis.

X-axis

The horizontal axis shows the **actual values** of the variable. The centre is μ .

Y-axis: normal PDF

The vertical axis shows **probability density**, written $f(x)$:

$$f(x) = 1/(\sigma\sqrt{2\pi}) \times \exp[-\frac{1}{2}((x-\mu)/\sigma)^2]$$

The PDF height is not $P(X=x)$. For a continuous variable, $P(X=x)=0$. The curve's area over an interval represents probability.

Mean, median and mode

$$\text{Mean} = \text{Median} = \text{Mode} = \mu$$

68–95–99.7 rule — precise rounded values

Interval	Approx. percentage
$\mu - \sigma$ to $\mu + \sigma$	68.27%
$\mu - 2\sigma$ to $\mu + 2\sigma$	95.45%
$\mu - 3\sigma$ to $\mu + 3\sigma$	99.73%

More precisely: 68.2689%, 95.4500%, and 99.7300%. The familiar 68%, 95%, and 99.7% are rounded labels.

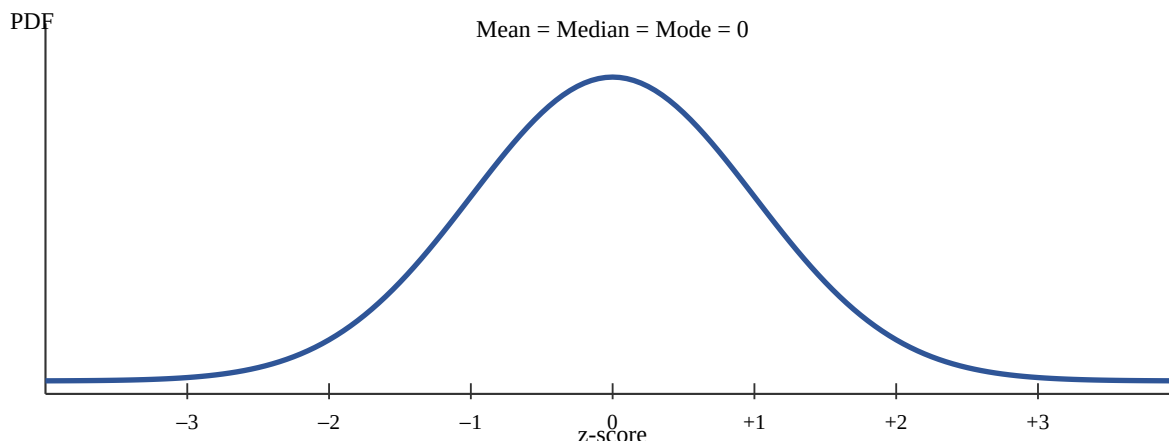
Measurements influenced by many small sources of variation can be approximately normal. Examples include adult height in suitable populations, measurement errors, some manufacturing dimensions, and some standardized test measurements. Not every real-world variable is normal.

3. Standard Normal Distribution

The standard normal distribution is the special normal distribution with:

$$\mu = 0, \quad \sigma = 1$$

$$Z \sim N(0, 1)$$



Graph note: The standard normal curve also approaches the x-axis in both tails but never touches it.

X-axis: z-score

The horizontal axis is usually called the **z-axis**. The z-score is:

$$z = (x - \mu)/\sigma$$

$z = 0$ means at the mean; $z = +1$ means one standard deviation above the mean; $z = -2$ means two standard deviations below the mean.

Y-axis: standard normal PDF

The vertical axis is still **probability density**. Since $\mu=0$ and $\sigma=1$:

$$\phi(z) = 1/\sqrt{2\pi} \times \exp(-z^2/2)$$

The maximum density occurs at $z=0$ and is approximately 0.3989. This is density, not point probability.

Mean, median and mode

$$\text{Mean} = \text{Median} = \text{Mode} = 0$$

68–95–99.7 rule for the standard normal

Interval	Approx. percentage
$-1 \leq Z \leq +1$	68.27%
$-2 \leq Z \leq +2$	95.45%
$-3 \leq Z \leq +3$	99.73%

4. Why Do We Need the Standard Normal?

There are countless normal distributions because μ and σ can vary. A common reference distribution lets us put all normal variables onto one universal scale.

$$z = (x - \mu)/\sigma$$

Example: if $\mu=70$, $\sigma=10$, and $x=80$:

$$z = (80 - 70)/10 = 1$$

Thus $x=80$ is one standard deviation above the mean. Once expressed as z , the standard normal reference distribution can be used.

5. Final Comparison

Feature	Normal	Standard normal
Notation	$X \sim N(\mu, \sigma^2)$	$Z \sim N(0, 1)$
Mean	μ	0
SD	σ	1
Horizontal axis	Actual value x	z -score
Vertical axis	PDF $f(x)$	PDF $\phi(z)$
PDF	$1/(\sigma\sqrt{2\pi}) \exp[-\frac{1}{2}((x-\mu)/\sigma)^2]$	$1/\sqrt{2\pi} \exp(-z^2/2)$
Mean = Median = Mode	μ	0

Remember: PMF → discrete point probabilities. PDF → continuous density. Normal → actual-value bell curve. Standard normal → bell curve on the z -score scale. The bridge is $z = (x-\mu)/\sigma$.